Ritesh Tiwari

riteshtiwari12.github.io | ritesh.tiwari.mathematics@gmail.com

ABOUT ME

Mathematics undergraduate at NIT Surat with research interests in real analysis, numerical computation, and mathematical visualization. Has a strong foundation in mathematical reasoning and programming, with a focus on applying theoretical concepts to computational and research-oriented projects.

EDUCATION

Integrated M.Sc. in Mathematics 2024–Present
Sardar Vallabhbhai National Institute of Technology Surat, Gujarat, India

- **Current CGPA**: 9.00
- **Related coursework**: Real Analysis, Calculus, Group Theory, Graph Theory, Data Structures.

CLASS XII [Board: CBSE] 2023
Kendriya Vidyalaya, NERIST, Nirjuli, Arunachal Pradesh

- **Percentage**: 92.4%

CLASS X [Board: CBSE] 2021
Vivekananda Kendra Vidyalaya, P.T.C Banderdewa, Arunachal Pradesh

- **Percentage**: 94.8%

TECHNICAL SKILLS

- **Core Strengths**: Logical reasoning, mathematical thinking, analytical problem-solving
- **Programming languages**: C++, Python, MATLAB
- **Mathematical tools**: LaTeX, Desmos
- **Python libraries**: Numpy, Matplotlib, Sympy (basic knowledge)
- **Other tools**: Jupyter Notebook, VS Code

COURSES & CERTIFICATIONS

- **Linear Algebra For Machine Learning [STTP]** – LDRP Institute of Technology & Research, Gujarat Technological University (December 01–05, 2025)
- **Mastering LaTeX & Scientific Sketching** – IMPA & Infinity Research and Development Institute (November 12–14, 2025)
- **Statistics & Data Analysis Workshop** – Department of Mathematics, Sikkim University (October 24–25, 2025)
- **SciComm 101 Workshop** – Office of Communication (OoC), IISc Bangalore (October 13–16, 2025)

- **Python for Linear Algebra and Numerical Methods** – IMPA & Infinity Research and Development Institute (July 28–30, 2025)
- **MATLAB Course** – Internshala (May–July 2025)
- **Single Variable Calculus –** MIT OpenCourseWare

PROJECTS

Solving Ordinary Differential Equations Using Numerical Methods in MATLAB (May–July 2025)

- Implemented Euler's, Heun's, and Runge-Kutta methods for solving ODEs.
- Compared results with analytical solutions and MATLAB's built-in function `ode45`.
- Created graphical visualizations to compare the precision and convergence of each method.

INTERESTS

- Mathematical visualization
- Solving mathematical puzzles